Evaluation runs: **commit** (s3-commit)

generated 2026-11-08 from brain-data 38584f536 2026-11-08 20:45:11 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): (the single pair): expected conforms. Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". G_{ref}, G_{cand} = the graphs of the reference and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is G_{ref}^{expl}); G_x = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

stateless

stateless: pair **single**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
2,0	4	4	conforms	0.0	4	1 h, 250 GB	4	10	brain03, 2026-11-02
2,1	4	4	conforms	0.0	4	1 h, 250 GB	4	10	brain03, 2026-11-02
2,2	4	4	conforms	0.0	2	1 h, 250 GB	4	10	brain03, 2026-11-02
4,0	16	16	conforms	0.0	2	1 h, 250 GB	16	136	brain03, 2026-11-02
4,2	16	16	conforms	0.0	4	1 h, 250 GB	16	136	brain03, 2026-11-02
4,4	16	16	conforms	0.0	4	1 h, 250 GB	16	136	brain03, 2026-11-02
8,0	256	256	conforms	1.5	6	1 h, 250 GB	256	32896	brain03, 2026-11-02
8,4	256	256	conforms	1.5	6	1 h, 250 GB	256	32896	brain03, 2026-11-02
8,8	256	256	conforms	1.1	6	1 h, 250 GB	256	32896	brain03, 2026-11-02
12,0	4096	4096	conforms	604.2	160	1 h, 250 GB	4096	8390656	brain03, 2026-11-02
12,6	4096	4096	conforms	678.9	82	1 h, 250 GB	4096	8390656	brain03, 2026-11-02
12,12	4096	4096	conforms	532.0	49	48 h, 250 GB	4096	8390656	brain04, 2026-11-03
16,0	65536		🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-02
16,16	65536		🕓 (48 h)			48 h, 250 GB			brain04, 2026-11-03

Not run (stateless). planned but never scheduled (grid points beyond the ladder): 2 4 fifo, 2 6 fifo, 2 8 fifo, 2 10 fifo, 2 12 fifo, 2 16 fifo, 2 20 fifo, 3 0 fifo, 3 1 fifo, 3 2 fifo, 3 4 fifo, 3 6 fifo, 3 8 fifo, 3 10 fifo, 3 12 fifo, 3 16 fifo, 3 20 fifo, 4 1 fifo, 4 6 fifo, 4 8 fifo, 4 10 fifo, 4 12 fifo, 4 16 fifo, 4 20 fifo, … (161 instances in all).

skipped: a smaller size had reached its limit: 16 1 fifo, 16 2 fifo, 16 8 fifo, 17 0 fifo, 17 1 fifo, 17 2 fifo, 17 8 fifo, 17 16 fifo, 18 0 fifo, 18 1 fifo, 18 2 fifo, 18 8 fifo, 18 16 fifo, 19 0 fifo, 19 1 fifo, 19 2 fifo, 19 8 fifo, 19 16 fifo, 20 0 fifo, 20 1 fifo, 20 2 fifo, 20 8 fifo, 20 10 fifo, 20 16 fifo, … (46 instances in all).

stateful

stateful: pair **single**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
2,0	4	4	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
2,1	4	4	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
2,2	4	4	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
4,0	16	16	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
4,2	16	16	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
4,4	16	16	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
8,0	256	256	conforms	0.1	4	1 h, 250 GB	brain02, 2026-11-02
8,4	256	256	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
8,8	256	256	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
12,0	4096	4096	conforms	1.7	32	1 h, 250 GB	brain02, 2026-11-02
12,6	4096	4096	conforms	1.7	36	1 h, 250 GB	brain02, 2026-11-02
12,12	4096	4096	conforms	1.9	34	48 h, 250 GB	brain04, 2026-11-03
16,0	65536	65536	conforms	1007.8	643	1 h, 250 GB	brain02, 2026-11-02
16,8	65536	65536	conforms	1013.1	643	1 h, 250 GB	brain02, 2026-11-02
16,16	65536	65536	conforms	958.0	643	48 h, 250 GB	brain04, 2026-11-03
20,0			(1 h)			1 h, 250 GB	brain02, 2026-11-02
20,20			(48 h)			48 h, 250 GB	brain04, 2026-11-04

Not run (stateful). planned but never scheduled (grid points beyond the ladder): 2 4 fifo, 2 6 fifo, 2 8 fifo, 2 10 fifo, 2 12 fifo, 2 16 fifo, 2 20 fifo, 3 0 fifo, 3 1 fifo, 3 2 fifo, 3 4 fifo, 3 6 fifo, 3 8 fifo, 3 10 fifo, 3 12 fifo, 3 16 fifo, 3 20 fifo, 4 1 fifo, 4 6 fifo, 4 8 fifo, 4 10 fifo, 4 12 fifo, 4 16 fifo, 4 20 fifo, … (185 instances in all).

skipped: a smaller size had reached its limit: 20 1 fifo, 20 2 fifo, 20 10 fifo, 21 0 fifo, 21 1 fifo, 21 2 fifo, 21 10 fifo, 21 20 fifo, 22 0 fifo, 22 1 fifo, 22 2 fifo, 22 10 fifo, 22 20 fifo, 23 0 fifo, 23 1 fifo, 23 2 fifo, 23 10 fifo, 23 20 fifo.

offline

offline: pair **single**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,0	4	4	conforms	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
2,1	4	4	conforms	0.1	8	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain01, 2026-11-02
2,2	4	4	conforms	0.0	8	10 min, 16 GB	0.0	0.0	0.0	in-memory	brain01, 2026-11-02
4,0	16	16	conforms	0.1	8	10 min, 16 GB	0.1	0.0	0.0	in-memory	brain01, 2026-11-02
4,2	16	16	conforms	0.1	8	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain01, 2026-11-02
4,4	16	16	conforms	0.0	8	10 min, 16 GB	0.0	0.0	0.0	in-memory	brain01, 2026-11-02
8,0	256	256	conforms	0.1	8	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain01, 2026-11-02
8,4	256	256	conforms	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
8,8	256	256	conforms	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
12,0	4096	4096	conforms	1.6	74	10 min, 16 GB	0.4	0.4	1.1	in-memory	brain01, 2026-11-02
12,6	4096	4096	conforms	1.5	74	10 min, 16 GB	0.4	0.4	1.1	in-memory	brain01, 2026-11-02
12,12	4096	4096	conforms	1.6	10	48 h, 250 GB	1.5	0.4	0.1	merge	brain02, 2026-11-08
16,0	65536	65536	conforms	12.7	155	1 h, 250 GB	9.0	9.1	3.5	merge	brain01, 2026-11-03
16,8	65536	65536	conforms	12.7	155	1 h, 250 GB	8.9	9.0	3.7	merge	brain01, 2026-11-03
16,16	65536	65536	conforms	12.4	156	1 h, 250 GB	5.4	8.9	3.5	merge	brain01, 2026-11-03
20,0	1048576	1048576	conforms	274.3	2378	1 h, 250 GB	206.9	206.2	67.5	merge	brain01, 2026-11-03
20,10	1048576	1048576	conforms	272.6	2375	1 h, 250 GB	205.3	204.7	67.3	merge	brain01, 2026-11-03
20,20	1048576	1048576	conforms	274.2	2379	1 h, 250 GB	111.5	206.9	67.3	merge	brain01, 2026-11-03

Not run (offline). planned but never scheduled (grid points beyond the ladder): 2 4 fifo, 2 6 fifo, 2 8 fifo, 2 10 fifo, 2 12 fifo, 2 16 fifo, 2 20 fifo, 3 0 fifo, 3 1 fifo, 3 2 fifo, 3 4 fifo, 3 6 fifo, 3 8 fifo, 3 10 fifo, 3 12 fifo, 3 16 fifo, 3 20 fifo, 4 1 fifo, 4 6 fifo, 4 8 fifo, 4 10 fifo, 4 12 fifo, 4 16 fifo, 4 20 fifo, … (204 instances in all).

the tool ended with an error (exit-127): 12 12 fifo.